



STATEMENT OF WORK

Voice AI Analytics

Audio Intelligence & Conversational Data Interface

Points of Contact

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1. Overview

Heya AI builds and operates AI-powered voice agents for business clients. These agents handle inbound and outbound calls including appointment booking, FAQ handling, and customer routing.

Each call produces a recording and a full transcript. The platform currently surfaces the following analytics to clients through a web dashboard:

- Total call volume
- User sentiment (positive / neutral / negative, sourced from Retell)
- Call disconnection reason
- Call success rate

These metrics are derived entirely from response metadata and transcript content. The call audio recordings are retained but are not currently analysed. This Statement of Work defines the scope of work to extend the platform with two capabilities: an audio intelligence layer and a conversational data interface.

2. Objectives

The Contractor is engaged to design and deliver the following:

1. An audio analysis pipeline that processes call recordings to extract behavioural and emotional signals not available from transcripts alone, and surfaces these as new metrics within the client dashboard.
2. A conversational interface embedded in the dashboard that allows business owners to query their call data using natural language, without requiring technical knowledge or manual data exports.

3. Scope of Work

3.1 Audio Intelligence Pipeline

The Contractor shall build a pipeline that ingests call audio recordings and produces structured, per-call insight data to be stored in a database. The following represent the target output categories. The Contractor may propose alternative or additional signals, provided they deliver demonstrable value to a business client reviewing their dashboard.

Target Audio Analytics	
Emotion Detection	Per-speaker emotion classification (e.g., happy, frustrated, neutral, anxious) at a turn or segment level, derived from acoustic features rather than transcript text.
Engagement Scoring	A composite metric reflecting the caller's level of active engagement, informed by speaking rate, response latency, and vocal energy across the duration of the call.
Sentiment Trajectory	A representation of how the caller's emotional state evolves over the course of the call capturing whether sentiment improves or deteriorates from opening to close.
Silence & Interruption Analysis	Detection of extended pauses, overlapping speech, and interruption events that may indicate friction, confusion, or disengagement.
Speaking Rate & Hesitation	Identification of moments where either party speaks at an atypical rate or displays hesitation markers, potentially signalling uncertainty or discomfort.
Acoustic Stress Indicators	Pitch variance and energy patterns associated with elevated emotional states, flagged at a call or segment level.
Agent Consistency Metrics	Cross-call comparison of agent vocal patterns tone, pace, clarity to surface performance variation across agents or time periods.

All generated insights must be persisted to the existing database, keyed to the corresponding call record, and exposed via API for consumption by the dashboard frontend.

3.2 Conversational Data Interface

The Contractor shall build a natural language query interface that enables a business owner to ask questions about their call data and receive accurate, data-grounded responses. The system must handle queries including but not limited to:

- Call volumes over specified time ranges
- Most frequently raised topics or questions within a given period
- Peak call times by hour or day of week
- Calls matching a specific topic, sentiment, or outcome
- Comparative analysis across agents or time periods

The recommended architecture is Retrieval-Augmented Generation (RAG): call transcripts, metadata, and audio-derived insights are embedded into a vector store, and an agent framework is used to route incoming queries directing content-based queries to semantic search and statistical or aggregation queries to structured database lookups.

The interface must be delivered as a REST API endpoint consumable by the existing dashboard frontend. Data access must be strictly scoped per client — the system must not return data belonging to other accounts.

4. Technical Environment

4.1 Existing Infrastructure

The Contractor will be provided access to the following:

- source of call recordings, transcripts, and post-call metadata which includes: call ID, timestamp, duration, transcript, payload (sentiment, disconnection reason, success flag), and a reference to the stored audio file

A dataset of anonymised call recordings and associated records will be provided for development and testing. This data must not be used for any purpose outside this engagement.

4.2 Technology

The Contractor is free to select the tools, frameworks, and models best suited to deliver the requirements. The following are provided as reference points only:

Reference Technologies	
Audio Processing	librosa, openSMILE, SpeechBrain, pyAudioAnalysis, OpenAI Whisper
Emotion / Speech Models	Hugging Face Transformers (wav2vec2, emotion2vec), pre-trained Speech Emotion Recognition (SER) models
Vector Store	Pinecone, Weaviate, Chroma, pgvector
RAG / Agent Framework	LangChain, LlamaIndex, or equivalent
LLM Provider	OpenAI, Anthropic Claude, or equivalent
Backend	Python preferred (FastAPI or Flask); other languages accepted with written justification

All third-party services, APIs, and models used must be documented, including licensing terms and any data residency or privacy implications.

5. Requirements

5.1 Functional Requirements

Audio Intelligence Pipeline

3. The pipeline must process a call recording and return a minimum of two distinct audio-derived insights per call.
4. All insights must be stored in the database, linked to the corresponding call record.
5. At least one insight must be surfaced in the client dashboard as a new or updated data visualisation.
6. The pipeline must support either near-real-time processing (within minutes of call completion) or scheduled batch processing; the chosen approach must be documented and justified.

Conversational Interface

7. The system must return accurate, data-grounded responses for the query types outlined in Section 3.2.
8. Responses must reference the underlying data where applicable (e.g., volumes, date ranges, relevant call counts).
9. The system must handle out-of-scope or unanswerable queries gracefully, without returning fabricated or hallucinated data.
10. Data access must be tenant-scoped — the system must not expose data across client accounts under any circumstances.

5.2 Non-Functional Requirements

- All handling of call recordings and personal data must comply with the Australian Privacy Principles (APPs) under the Privacy Act 1988. Raw audio must not be retained beyond what is operationally necessary for analysis.
- Conversational interface responses must be returned within 10 seconds under normal operating conditions.
- The codebase must include a README sufficient for a technical team member to deploy and operate the system without additional instruction.
- All external APIs, models, and data dependencies must be identified and documented.

6. Deliverables

The following deliverables are required under this engagement:

Deliverable Schedule	
Technical Design Document	A written document covering system architecture, data flow, model selection rationale, RAG design decisions, and any database schema changes. To be submitted and approved before build commencement.
Audio Intelligence Pipeline	A fully implemented, tested, and documented pipeline that processes call recordings and writes derived insights to the database.
Conversational Interface API	A deployed REST API endpoint implementing the RAG-based query system, with per-client data scoping enforced.
Dashboard	Frontend components required to surface new audio insights and the conversational interface within the existing dashboard.
Source Code Repository	A clean Git repository containing all code, dependency manifests, configuration, and a deployment README.
Handover Presentation	A walkthrough of the delivered system for the Heya AI technical team, covering architecture, key decisions, known limitations, and recommended next steps.

7. Assumptions & Dependencies

This SOW is issued on the following assumptions. Material changes to these assumptions may require a scope and timeline review.

- Heya AI will provide the Contractor with a complete anonymised dataset of call recordings and associated database records prior to commencement of build work.
- Heya AI will provide documentation of the existing database schema, API response structure.
- The Contractor is responsible for provisioning any third-party API access (LLM providers, vector databases, etc.) required to complete the work, unless otherwise agreed in writing.
- Heya AI will make a technical representative available to respond to clarification questions within two business days.